

Nifty 100 Financial Analytics & Valuation Platform

Comprehensive Enterprise Analyst Manual & System Guide

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

1. Architecture & ETL Data Pipelines

Details on pandas ingestion, normalization engines, and SQLite storage schema.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

2. Financial Ratio Engine Specifications

Mathematical definitions for ROE, ROCE, D/E, ICR, OPM, and 5Y CAGR computations.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

3. Data Quality & Validation Framework

14-rule assertion suite, severity logging, and winsorization thresholds.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

4. Streamlit Dashboard User Navigation

Step-by-step usage guide for Screener, Company Profile, Sector Analysis, and Peer Comparison.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

5. Cash Flow Intelligence Engine

CFO Quality scores, CapEx intensity classifications, and 8 capital allocation patterns.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

6. KMeans Clustering & Machine Learning

KMeans setup (k=5), feature scaling, centroid Euclidean distances, and 5 financial archetypes.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

7. PDF Reporting & Tearsheet Generation

ReportLab document templates, two-page tearsheets, dynamic charts, and batch PDF engines.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

8. FastAPI REST API Documentation

Guide to all 16 REST endpoints, query filters, JSON schemas, and OpenAPI specification.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

9. Pytest Suite & Test Automation

146-test coverage summary across ETL, ratios, rules, and FastAPI integration endpoints.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)

10. Troubleshooting & Deployment Guide

Server deployment, environment variables, load testing, and maintenance routines.

This section documents operational workflows, API interfaces, mathematical rules, and platform usage guidelines for financial analysts and system operators working with the Nifty 100 dataset.

Parameter	Specification / Value
Target Universe	Nifty 100 Listed Companies
Data Storage	Local CSV / SQLite Database
Backend Framework	FastAPI 1.0 (Python 3.13)
Testing Suite	146 Passing Pytest Cases (0 Failures)